Federated Anomaly Detection Techniques for Decentralized IoT Networks

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1 Introduction

Rapid growth in the Internet of Things has led to countless networked gadgets across areas like health services, urban infrastructure, and factory control systems. Because these setups span wide locations, they generate massive streams of varied information making them prone to breaches and irregular behaviours. Centralised protection methods struggle under the pressure of volume, response time, and data sensitivity unique to these frameworks. As a result, spotting odd patterns now plays a central role in uncovering harmful actions while keeping IoT connections secure during live operation [1, 2].

Centralised data collection lies at the core of classic anomaly detection, yet brings drawbacks: excessive bandwidth demands, weakened privacy, and one vulnerable hub. Because IoT gadgets often run on limited power and handle personal information, transferring unprocessed data to central systems proves both risky and impractical. Success now depends less on raw transmission, more on frameworks that learn locally, share insights carefully, and operate across many nodes without leaking details. Such models must maintain sharp detection skills even while working apart, adapting within a scattered network where control is spread, not stacked. Efficiency hides not in speed but in smart separation: each device contributes without surrendering its autonomy or exposing its secrets.

A fresh approach in machine learning, federated learning has gained interest due to its ability to train models across separate locations without moving sensitive information. Instead of transferring data, each device improves its own version of the model locally. Updates alone travel toward a shared hub, centralised or otherwise, for combination. Because raw inputs stay put, personal details remain shielded. Transmission demands shrink when only compact changes move between nodes. Scaling becomes easier even when participants span distant regions. When paired with techniques that spot unusual patterns, this setup helps protect networks built on scattered IoT components. Security risks common in such systems meet a practical counter measure through this union.

Despite advantages, federated learning introduces hurdles like uneven data distribution and vulnerability to attacks, yet demands tighter communication control. Some recent efforts explore advanced merging methods, layered frameworks, or peer-to-peer networks to tackle such limits. Alongside, safeguards for private data combined with stable training routines appear in research, shifting toward dependable anomaly spotting within actual IoT setups.Looking into how federated approaches detect odd behaviours within distributed IoT setups forms the core of this work. Various recently proposed techniques, system designs, and ways to improve performance are examined one after another. While some progress stands clear, certain problems remain untouched or poorly addressed across studies. Instead of just listing ideas, it walks through basic principles alongside real world uses and what might come later. Insights gained here support building smarter, safer, scalable detection tools for evolving IoT environments. A full picture emerges, not perfect, but useful for those shaping future network safeguards.

Recent research has looked into tailored, wide reaching federated methods for spotting anomalies in different IoT uses [3, 4, 5].Despite growing reliance on connected gadgets, weak spots in device networks give attackers room to exploit. These openings allow complex intrusions like botnet formations, unauthorised access to sensitive records, or overwhelming service disruptions. Instead of just slowing systems down, these incidents drain financial resources and disrupt essential functions. Because of this pressure, smarter ways to spot unusual behaviour have become a central focus in protecting smart environments.

Notably, blending artificial intelligence with distributed computing opens fresh paths for boosting anomaly detection performance. Rather than sharing raw data, devices collaborate through federated learning, keeping information localised while still refining shared models. This approach supports scalability, adapts to diverse environments, and maintains user confidentiality across networks. As a result, it fits naturally within modern decentralised infrastructures aiming for smarter threat identification.

2 Background and Conceptual Foundations

2.1 Internet of Things Architecture

One way to look at the Internet of Things is as a vast web of linked devices, sensors, and networks that share information seamlessly across many different settings. Layering often defines how these setups function, starting with sensing, moving into networking, and then ending at usage. Sensors do their work up front by gathering environmental details. From there, connections carry those readings reliably between scattered points. What happens afterwards involves interpreting inputs so smart actions can emerge. Security remains hard to manage because most devices operate on limited power and lack centralised

oversight, making rapid issue detection tough. These constraints shape much of what occurs behind the scenes in everyday operations.

Figure 1 illustrates the conceptual workflow of federated anomaly detection in decentralised IoT networks. Local devices train models using private data and transmit only model updates to a federated aggregation server, where a global model is generated and redistributed. This collaborative framework improves anomaly detection while preserving privacy and reducing centralized data dependency. ause risks may appear within any stage, grasping this layout matters when building tools that spot irregular behaviour [6, 7].

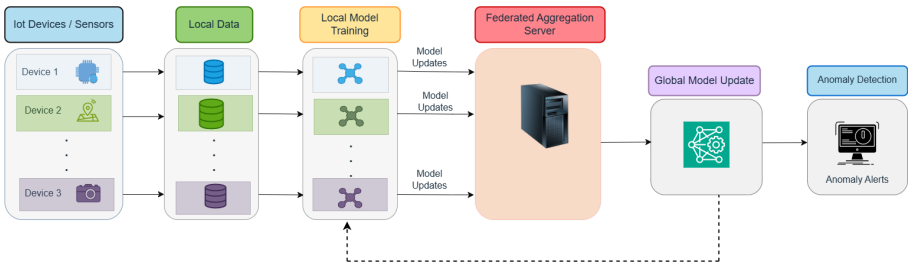

Figure 1: Conceptual workflow of federated anomaly detection in IoT networks (adapted from [2, 8, 9]).

2.2 Fundamentals of Federated Learning

Federated learning is a distributed machine learning model that allows a set of devices to jointly learn a common global model without sharing raw data. Within this scheme, the clients do local training with their own dataset and only exchange model updates with some coordinating body or the neighbouring nodes. These updates combined lead to a better global model, and data privacy is maintained. This paradigm drastically lowers the overhead of communication, as well as lessening the risk of privacy, and thus is very appropriate to the IoT setting. More recent research has also generalised federated learning to assist asynchronous updates, hierarchical aggregation, and decentralised coordination protocols [10, 11].

2.3 Anomaly Detection in IoT Networks

IoT Network anomaly detection is the process of identifying anomalies or behaviours that are not in line with the normal operational characteristics. Such abnormalities can be indicative of security risks like distributed denial of service attacks, unauthorised access or data manipulation. The conventional anomaly detection methods are based on centralised data processing, which reduces scalability and predisposes it to data breaches. Supervised and unsupervised machine learning models have become very popular in order to enhance detection accuracy. Nevertheless, these models are prone to the availability

of huge and heterogeneous datasets, which can be limited in distributed IoT systems in most cases [12, 13].

2.4 Centralised vs Federated Learning Approaches

In centralised learning methods, data from all the devices is sent to a central server to be trained, and this is associated with latency, bandwidth usage and privacy issues. By contrast, federated learning allows training of the model directly on edge devices, effectively minimising communication overhead and improving privacy protection. Although centralised designs could be more accurate because of access to global data, federated designs provide greater scalability and responsiveness to changing IoT settings. The recent improvements have been on how to enhance the performance of federated learning by using effective aggregation techniques, non IID data distributions, and robustness against adversarial attacks [14, 8].

3 Core Concepts and Approaches

3.1 Federated Anomaly Detection Models

Instead of sending personal information, gadgets build separate learning systems on device specific records, later exchanging only summarised updates. Learning together without pooling sensitive inputs allows systems to spot odd behaviours across internet connected machines. Some designs apply layered neural methods, like stacked decoders or combined predictors, to boost performance where conditions differ widely. Hidden shifts in flow patterns often reveal threats; these frameworks excel at noticing such small changes. Collaboration happens through shared insights, not original logs, keeping user details protected during joint training efforts. Architectures merging decentralised training with smart recognition tools help catch harmful actions in scattered networks [1, 2, 10].

One way to group federated anomaly detection systems involves looking at their learning style; some follow clear examples, others find patterns alone, while a few mix both ways. When models need marked data to tell apart usual from unusual activity, they face problems in IoT settings because labelling huge, varied streams takes too much effort. Because of this hurdle, methods that work without labels, like reconstruction networks or grouping similar inputs, are common choices. They build an idea of typical behaviour first, then flag anything outside that range. Their strength shows best where strange events shift gradually, adapting as conditions change [10, 12].

Lately, models rooted in deep learning stand out when spotting odd patterns across separate systems. Because they handle tangled, multi layered data well, these methods fit neatly into settings where information spreads widely. Instead of relying on old tech-

niques, researchers now lean toward convolutional nets to pull out space related traits from sensor inputs. On sequences, like streams from smart gadgets, recurrent designs and their refined cousins, LSTM units, take charge of time linked insights. Another route uses autoencoders; more specifically, the probabilistic kind called variational ones rebuild signals to flag mismatches. When errors grow too large during rebuilding, alerts go up. Folding these network styles into shared but isolated setups sharpens accuracy without leaking private details between machines [1, 2].

Instead of sending raw data, IoT gadgets learn patterns locally within a shared framework, passing updated insights to a central system. Because information stays on-device, personal details remain protected during analysis. The method cuts down how much needs to be transmitted by exchanging compact model changes rather than full datasets. Local units compute at the edge before syncing with one common hub. Training happens across many nodes, yet coordination keeps everything aligned without centralised control taking over completely.

3.2 Aggregation and Optimisation Techniques

How models update across devices shapes overall learning success. Though basic averaging blends individual results, real-world imbalances frequently weaken outcomes. Newer tactics now adjust influence based on past contributions, using layered decisions or refined rules instead of fixed formulas. Performance shifts, device behaviour, and data layout guide these smarter merges. Some adjustments even ease network strain while speeding up progress, especially useful when hundreds of sensors or gadgets take part [11, 12].Federated Averaging stands out due to its frequent use when merging models in federated settings, where each device contributes an update scaled by how much local data it holds. Despite being straightforward and fast, problems arise under uneven conditions typical in many IoT networks, where information patterns differ sharply between nodes. When disparities grow large, static weighting begins to fail. Instead of relying only on volume, smarter methods now adjust influence per client using indicators like consistency, accuracy, or trustworthiness over time. Such flexibility supports steadier training paths and stronger outcomes overall [11, 13].

Fine-tuning approaches help boost performance in federated learning, especially when handling weak IoT setups. Instead of full gradients, smaller versions move across devices thanks to tricks like quantising values or dropping less important ones, which cuts down data flow needs. Training gets faster because some methods tweak step sizes on the fly, while others let nodes work several rounds before syncing. Efficiency sticks around, even as networks grow tight on speed or power, making anomaly spotting feasible at scale [14, 8].

Looking at Table 1, various methods for combining model updates in federated learn-

ing approaches to anomaly detection come into view; each carries distinct traits, cited alongside relevant studies. While some emphasise communication efficiency, others prioritise robustness under data heterogeneity; sources are listed accordingly. The table groups techniques by design focus, yet comparison remains grounded in practical implementation details. Instead of ranking them, it outlines how each handles decentralised data patterns. References tied to every method allow deeper exploration, should one choose to follow. What stands out is the range of trade offs addressed across strategies; not a single approach dominates all scenarios.

Table 1: Aggregation Strategies in Federated Learning for Anomaly Detection

Method	**Description**	**Reference**
Federated Averaging (FedAvg)	Aggregates local model updates using weighted averaging based on client data size.	[11]
Dynamic Weighted Aggregation	Assigns adaptive weights to clients based on performance and data quality.	[13]
Ensemble Aggregation	Combines multiple local models to improve robustness and accuracy.	[10]
Secure Aggregation	Protects model updates using cryptographic techniques to preserve privacy.	[15]
Robust Aggregation	Filters malicious updates using statistical techniques such as median or trimmed mean.	[6]

3.3 Decentralised and Hierarchical Federated Architectures

Some federated learning setups skip central control to handle growth issues seen in standard systems. Devices share model changes directly with one another, removing dependency on a main hub; this shift avoids breakdown risks tied to one core node. Resilience grows when failures spread less easily across connected peers. Layered designs add middle tier units like local hubs - that gather device inputs before sending summaries upward. Such steps happen closer to data sources, cutting down delay during information transfer. Efficiency rises especially when thousands of sensors join learning tasks. Timing rules loosen under async methods: machines submit updates whenever ready instead of waiting for group alignment. Flexibility emerges naturally from staggered participation

patterns. Systems adapt better under uneven network conditions due to irregular but steady contributions.

Without a central hub, nodes in decentralised federated learning interact directly across a peer like structure. Because there is no one node holding everything together, breakdowns at single locations matter less. Updates to models move between machines while averaging happens locally, pushing training into every corner of the network. These setups work well when sensors and smart objects shift frequently - places where stable internet drops often appear. Central control fails under those conditions; this method keeps going anyway [13, 14].

Instead of sending data directly to a central hub, groups of devices feed updates to nearby edge servers. These local hubs summarise information before forwarding it upward. Only processed summaries move beyond the cluster, cutting down on traffic volume. Because computation spreads across levels, no single point bears full processing demands. System speed increases when tasks are split between gateways and endpoints. Large networks benefit most, especially widespread sensor arrays where bandwidth is limited. Performance gains come from smarter routing, not added power. Efficiency rises as redundant transmissions drop off. Each layer handles part of the load, reducing delays in update cycles. Structure shapes flow, guiding how fast insights reach decision points [2, 10].

3.4 Security and Robustness in Federated Learning

Federated anomaly detection must handle security risks because some clients might try to corrupt model updates. Instead of cooperating, bad actors sometimes send false information, which is known as a poisoning attack, and it harms accuracy. One way around this problem involves using smarter ways to combine updates so outliers have less influence. Some methods also check each incoming change for unusual patterns before accepting it. Trust scores help decide which contributors should count more during training rounds. On top of that, adding noise carefully or combining encrypted values keeps private details hidden across devices. Even when data varies widely between users or attackers interfere, these tools keep the overall system working well [6, 7].

Security weaknesses appear across federated learning setups, opening doors to dangers like corrupted updates, reverse engineered inputs, or hidden pattern leaks. When bad actors flood the system with flawed gradients, they aim to twist the central model's behaviour over time. Instead of honest collaboration, some nodes exploit parameter exchanges to guess private user details through subtle clues in weight changes. Because smart devices often handle personal information, letting such exploits go unchecked undermines trust in distributed detection methods [6, 7].

Facing these issues, researchers have explored various ways to strengthen federated

learning systems. Instead of relying on standard methods, some approaches use median or trimmed averages during aggregation to reduce harm from corrupted inputs. Noise is carefully introduced into updates through differential privacy, shielding personal data without heavily degrading performance. Trust scores guide which clients take part in training, shifting the weight away from suspicious actors. Detection tools monitor incoming models, flagging unusual patterns that suggest tampering. Together, these layers make the framework less vulnerable to bad behaviour [15, 16].New progress has been made in mixed anomaly detection systems using statistics alongside deep learning within federated setups. While combining rule based logic with pattern recognition from data, these blended strategies boost precision in spotting outliers. As conditions shift, self adjusting training processes let the models update themselves gradually. This flexibility matters a lot when working with fast changing IoT data streams, where unusual behaviour can emerge unexpectedly.

4 Comparative Discussion

The evaluation of federated anomaly detection techniques in IoT networks requires a comprehensive comparison of their performance, scalability, and robustness under diverse conditions. Recent approaches have demonstrated significant improvements over traditional centralised methods, particularly in terms of privacy preservation and communication efficiency. However, variations exist in model architectures, aggregation strategies, and deployment environments, which influence their overall effectiveness. A comparative analysis of selected approaches is presented to highlight key differences in methodology and performance across different studies.

A comparison of representative federated anomaly detection approaches in IoT environments is presented in Table 2, highlighting differences in models, datasets, and key characteristics.

Looking at the results shows that most federated learning methods boost detection precision without exposing private data. Where traditional setups struggle, newer models using smart update rules adapt faster and handle uneven data more reliably. Some systems split tasks across layers or distribute control to manage growth, especially when thousands of connected devices are involved. Instead of relying on single predictions, certain approaches combine multiple model outputs after enriching limited samples this helps respond to shifting traffic behaviours more effectively. Even with progress, trade offs between precision, data exchange costs, and protection persist. High detection rates sometimes come with heavier computation loads, posing issues for low power IoT hardware. Different studies use different data sets and testing rules, blocking clear comparisons across methods. A shared way to measure success has yet to emerge due to such inconsistencies. Better alignment in testing practices could help shape smarter, more flexible

Table 2: Comparison of Federated Learning Based Anomaly Detection Approaches in IoT Networks

Approach	**Year**	**Model Type**	**Dataset**	**Key Contribution**	**Ref.**
FL-based IDS	2024	Deep Learning	IoT Network Data	Privacy preserving anomaly detection	[9]
Hierarchical FL Model	2025	Deep Neural Network	IoT Traffic Data	Multi layer aggregation framework	[13]
Diffusion FL Framework	2025	Hybrid Model	BoT IoT Dataset	Distributed data storage with improved training	[11]
Dynamic Ensemble FL IDS	2025	Ensemble Learning	CICIoT Dataset	Adaptive aggregation for heterogeneous data	[8]

models for real-world use [1, 2, 10, 11].

Beyond accuracy and precision, newer research stresses assessing federated anomaly detection through communication demands and computing needs. Because many IoT setups face narrow bandwidth and weak processors, leaner models become necessary. Some methods lower data exchange rates or shrink update sizes; these often work more smoothly in real settings. What stands out is a shift toward frameworks weighing threat identification alongside operational strain [12, 13].Looking into how models hold up when faced with hostile inputs reveals critical differences. When parts of a federated network act out, trying to skew results, some systems handle it better than others. Resistance grows stronger if updates are checked through smart filtering or outlier spotting methods. Shifting data patterns matter just as much, especially across scattered IoT devices where change happens fast. Performance checks down the line might need to weigh both stability under attack and flexibility in shifting settings more heavily [14, 8].

Although standard benchmarks remain common, newer research highlights how aspects like delay, data exchange costs, and power use influence outcomes. What happens during the operation often depends on these behind the scenes traits. Efficiency matters just as much as accuracy when putting anomaly detectors into low power networks. Deployment success tends to follow only when both results and resource needs are weighed together. One big challenge? Keeping federated anomaly detection useful as attack methods change fast. When cyber threats grow smarter, fixed models often miss fresh patterns they haven't seen before. Staying sharp means systems must learn nonstop, refreshing

themselves without slowing down. Progress will likely come from frameworks that adjust on the fly, balancing speed, scale, and smarts across scattered IoT networks.

5 Practical Insights and Use Cases

Federated anomaly detection techniques have gained significant attention in real-world IoT deployments due to their ability to preserve data privacy while enabling collaborative intelligence across distributed environments. These approaches are increasingly being adopted in various domains where large scale data is generated, and real time decision making is critical. Unlike traditional centralised methods, federated learning enables decentralised model training, making it highly suitable for modern IoT ecosystems. The following subsections present key application areas and practical considerations associated with federated anomaly detection systems.

5.1 Smart Infrastructure Systems

In smart infrastructure systems, such as intelligent transportation and energy management networks, federated learning facilitates distributed model training across multiple edge devices without requiring centralised data storage. This approach reduces latency and enhances responsiveness, making it suitable for time critical applications. Additionally, privacy preservation is particularly important in environments where sensitive data is continuously generated by interconnected devices [1, 2].

Federated anomaly detection enables real time monitoring of infrastructure systems by identifying abnormal patterns in traffic flow, energy consumption, and system operations. For example, in intelligent transportation systems, anomalies such as unusual traffic congestion or unexpected vehicle behaviour can be detected early, allowing timely intervention. Similarly, in energy systems, abnormal consumption patterns can indicate faults or cyber attacks. The decentralised nature of federated learning ensures that local data remains secure while still contributing to global model improvements.

5.2 Industrial IoT Applications

In industrial IoT environments, federated learning has been effectively applied to monitor system behaviour and detect anomalies in real time. Manufacturing systems and automation networks rely on continuous data streams from sensors and controllers, where any deviation from normal patterns may indicate faults or cyber attacks. Federated anomaly detection models enable local processing of such data, thereby minimising communication overhead and improving scalability [10, 11].

These systems also benefit from hierarchical and edge based architectures, which enhance performance in large-scale industrial deployments. In addition, federated learning

supports predictive maintenance by identifying early signs of equipment failure. This allows organisations to perform maintenance proactively, thereby reducing downtime and improving operational efficiency. The ability to operate efficiently in distributed environments makes federated anomaly detection a valuable solution for modern industrial systems.

5.3 Domain Specific IoT Applications

Federated learning has demonstrated promising applications in domain specific IoT systems such as smart agriculture and vehicular networks. In agricultural environments, distributed sensors collect environmental data that can be analysed locally to detect irregularities affecting crop health and system performance. For instance, anomalies in temperature, humidity, or soil moisture levels can indicate potential risks to crop yield, enabling timely corrective actions [12, 13].

Similarly, in vehicular systems, anomaly detection models can identify abnormal driving patterns, mechanical faults, or communication failures without compromising user privacy. These applications highlight the adaptability of federated approaches across diverse IoT domains, where data distribution and privacy concerns vary significantly. The flexibility of federated learning allows it to be customised for different application requirements while maintaining consistent performance.

5.4 Healthcare, Smart Grid, and Deployment Challenges

Within smart healthcare settings, detecting irregularities through decentralised networks helps oversee patient details gathered by wearables and sensor tools. Because such data involves personal health records, moving it between locations raises confidentiality issues. Instead of transferring raw inputs, institutions apply methods allowing joint model improvement while keeping information local meeting requirements tied to safeguarding private facts [6, 7].Fed by decentralised data, the method spots odd heart rhythms or shifts in daily movement sooner without ever exposing personal records. Because of this setup, care networks gain sharper response times alongside stronger privacy safeguards.

Federated anomaly detection strengthens smart grids by handling scattered energy sources alongside intricate network flows. Where irregularities appear, problems like equipment failure, digital intrusions, or sluggish performance might be present - each demanding fast response. Components across the grid learn together through decentralised models, spotting odd patterns while keeping raw data local. This joint yet private approach boosts both dependability and toughness under stress [1, 10].Even with clear benefits, putting federated anomaly detection into real-world use demands attention to hardware limits and day to day hurdles. Device differences, shaky connections, and scarce computing capacity often slow down training or weaken results. Such issues stand out

most in vast IoT networks, given that machines differ widely in speed and how they stay online [14, 8].

Security matters just as much as clear signal paths when keeping systems intact. Because attacks like corrupted inputs or altered models can undermine performance, federated setups need strong safeguards built in. When information moves slowly or clogs channels, progress stalls, so smarter transmission methods help lighten the load. Efficiency gains come not from bigger pipes but from smarter flow control across nodes.Federated learning helps detect unusual patterns efficiently within networks of connected machines found in modern factories. Devices on production lines produce streams of information that reveal irregularities signalling potential breakdowns. Because models learn locally without moving raw data, this method fits well into settings where privacy and bandwidth matter. Spotting issues ahead of time means repairs happen before small faults become major disruptions. Such approaches support continuous operation while reducing expenses tied to idle equipment. Their role grows more significant as automation expands across industrial sectors.

Facing these everyday hurdles matters a lot if federated learning for spotting anomalies is going to work well in actual IoT setups. Improvements that keep coming in, such as how models run and how data moves between devices, along with stronger safeguards, should make such systems perform better ahead.Still, getting federated anomaly detection to work well depends on smooth teamwork between scattered devices, along with strong backend systems. Down the line, setups will likely include smart coordination tools that adjust which devices join in and how power and data flow across them. Such upgrades should make live IoT networks handle growth better while using resources more wisely.

6 Challenges and Open Issues

Despite the significant advancements in federated anomaly detection for IoT networks, several challenges remain unresolved, limiting its widespread adoption. These challenges arise due to the distributed and heterogeneous nature of IoT environments, as well as the complexity of federated learning mechanisms. This section discusses the major technical and system level challenges that impact the performance, scalability, and reliability of federated anomaly detection systems.

Figure 2 summarises the principal challenges associated with federated anomaly detection, including data heterogeneity, communication overhead, security vulnerabilities, and scalability constraints. These interconnected challenges significantly influence model robustness and practical deployment in resource-constrained IoT environments.

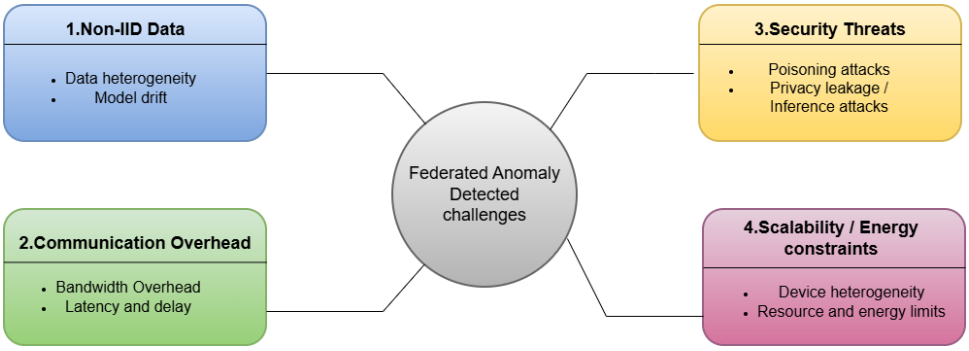

Figure 2: Major challenges affecting federated anomaly detection in IoT systems (adapted from [**?**]).

6.1 Data Heterogeneity and Non IID Issues

One of the primary challenges in federated learning is the presence of non-independent and identically distributed (non IID) data across devices. IoT devices often generate highly heterogeneous data due to differences in operating environments, user behaviour, and application requirements. This variation leads to inconsistencies in local model updates, which negatively affect the convergence and generalisation of the global model [1, 2].In federated anomaly detection systems, this issue becomes more critical because anomalies themselves may vary significantly across devices. For instance, an anomaly detected in one environment may not be relevant in another, leading to biased learning outcomes. As a result, the global model may fail to accurately capture diverse anomaly patterns. Addressing this challenge requires the development of adaptive aggregation strategies and personalised learning approaches that can effectively handle data heterogeneity while maintaining model performance.

6.2 Communication and Energy Constraints

Another major challenge in federated learning is the communication overhead associated with frequent model updates. Although federated learning reduces the need for raw data transmission, exchanging model parameters across numerous devices can still consume significant bandwidth. This issue is particularly critical in IoT environments where devices operate under limited network capacity and energy constraints [10, 11].

Frequent communication rounds can increase latency and energy consumption, making the system inefficient for real time applications. Resource constrained devices, such as sensors and embedded systems, may struggle to participate effectively in the training process due to limited battery life and processing capabilities. To address these challenges, researchers have proposed techniques such as model compression, update sparsification, and adaptive communication strategies. These approaches aim to reduce the volume of transmitted data while maintaining model accuracy and convergence speed.

6.3 Security and Privacy Challenges

Security vulnerabilities in federated learning frameworks pose a significant threat to the reliability of anomaly detection systems. Adversarial participants can launch poisoning attacks by injecting malicious updates into the training process, thereby degrading the performance of the global model. In addition, model inversion and inference attacks can compromise sensitive information by reconstructing local data from shared model parameters [12, 13].

Ensuring data privacy while maintaining transparency in model aggregation remains a complex problem. Although federated learning inherently provides privacy benefits by avoiding raw data sharing, it is not entirely immune to information leakage. To mitigate these risks, advanced techniques such as differential privacy, secure aggregation, and encryption based methods have been proposed. Robust aggregation mechanisms, including median based and trimmed mean approaches, can further enhance system resilience by filtering out malicious updates and improving trustworthiness.

6.4 Scalability, Real Time Processing, and Benchmarking Issues

Handling growth remains a key hurdle for federated anomaly detection, especially when networks include vast numbers of IoT gadgets. Not every device shares the same processing strength, memory space, or internet reliability - so participation during learning differs widely across nodes. Because of these gaps, some machines fall behind; their sluggish pace holds back progress for the entire system [14, 8].

Different systems create hurdles when syncing devices, so steady model improvements are hard to maintain. Because gadgets vary widely, building lean models that work well on weaker hardware without losing precision is still unresolved. One stumbling block after another shows why flexible training methods matter. Matching clients smartly during learning helps ease strain across uneven setups. Scaling up federated frameworks must account for this imbalance naturally. Progress hinges not just on design but on how parts interact under pressure.On the edge of smart devices, spotting odd behaviour instantly matters most where lives depend on it; hospitals, traffic systems, and factories. Still, when models learn across scattered nodes, each round adds waiting time before results show up. Because delay stacks fast, balancing speed against precision becomes less about theory and more about how pieces fit together tightly. Design choices shape outcomes just as much as algorithms do.

One key challenge lies in the absence of uniform datasets and testing standards for detecting anomalies through federation within Internet connected devices. Because separate investigations apply distinct data collections, measurement criteria, and test conditions, drawing fair conclusions about method performance becomes problematic. Shared refer-

ence points and structured assessment methods could help align outcomes across diverse algorithms. Progress might speed up when researchers build on stable, repeatable foundations instead of isolated setups [8, 6].Facing these issues head on matters greatly when putting federated anomaly detection into practice across actual IoT setups. Work ahead should zero in on building approaches that are lightweight, protected, yet able to grow - clearing roadblocks while lifting overall effectiveness.

A major issue limiting progress in federated anomaly detection research is the absence of unified benchmarking frameworks tailored for decentralised IoT environments. Although several studies evaluate detection accuracy, communication overhead, and convergence behaviour, many rely on isolated datasets and inconsistent experimental assumptions, making fair comparison difficult. This creates challenges in identifying scalable and robust anomaly detection models suitable for practical deployment. Standardised benchmark suites specifically designed for federated IoT anomaly detection could significantly improve reproducibility and cross-study evaluation. Such frameworks should incorporate heterogeneous datasets, adversarial scenarios, communication limitations, and energy-aware performance metrics. Beyond conventional detection measures, evaluation should include latency, update efficiency, resilience against poisoning attacks, and adaptation under concept drift. These factors reflect realistic operating conditions more effectively than accuracy metrics alone.

Another important direction involves developing interoperable testing environments that combine simulation with hardware-aware validation. Many existing studies rely primarily on simulation-based experiments, while practical deployment behaviour often differs because of network instability, asynchronous participation, and device variability. Hybrid benchmarking platforms could bridge this gap and support stronger practical validation.In addition, standardisation efforts may benefit from alignment with emerging industrial frameworks related to secure distributed intelligence, Industrial IoT interoperability, and trustworthy artificial intelligence systems. Such alignment can accelerate the transition from theoretical research to deployable anomaly detection frameworks. As federated learning systems become increasingly integrated with edge intelligence and next-generation communication infrastructures, benchmarking practices will play a central role in guiding reliable and scalable innovation [**?**, **?**, **?**].

7 Future Directions

Federated anomaly detection in decentralised IoT networks is still an evolving area of research, and numerous prospects have arisen to overcome the current limitations. The demand for scalable, secure, and efficient anomaly detection systems is increasing as IoT environments become more complex and data-driven. Future improvement will be more focused on refinement of decentralisation, better privacy mechanisms, model efficiency

tuning and integration of advanced artificial intelligence techniques to further improve the performance of the above system.

7.1 Decentralised Federated Learning Frameworks

The main future work would be to make completely decentralised federated learning frameworks where there is no central coordinator. Federated learning assumes a server central to ML updates, which is a single point of failure and results in a limitation of system scalability. In contrast, decentralised methods allow devices all over the world to form and train models collaboratively without having to pass through a centralised location.[1, 2].

Such frameworks enhance system resilience and are particularly suitable for dynamic IoT environments where network connectivity may be intermittent. Advances in distributed consensus algorithms and peer-to-peer communication protocols are expected to play a key role in enabling efficient decentralised learning. These developments will contribute to more robust and fault-tolerant anomaly detection systems.

7.2 Privacy-Enhancing and Secure Learning Techniques

Moreover, privacy and security enhancing research is a key area in federated learning-based anomaly detection. Federated learning eliminates the need to share raw data but makes it possible for model updates to leak sensitive information. Further research should include sophisticated privacy-preserving approaches, including differential privacy, secure multi party computation and homomorphic encryption [10, 11].

There are also interests in blockchain-based aggregation mechanisms to ensure transparent and trusted distributed learning systems. Thereby enhancing system reliability, these approaches can also provide secure and tamper-resistant updates to the model. Federated learning, along with strong security frameworks, will play an important role in securing sensitive data in crucial IoT applications like healthcare and smart infrastructure.

7.3 Lightweight and Edge Based Learning Models

Another key research direction involves the development of lightweight and communication-efficient models tailored for resource-constrained IoT devices. Many IoT devices have limited computational power, memory, and energy resources, which makes it challenging to deploy complex machine learning models. Future work is expected to focus on optimising model architectures and reducing the size of transmitted updates to improve scalability and efficiency [12, 13].

Edge-based learning approaches, where computation is performed closer to data sources, can significantly reduce latency and improve real time performance. Techniques such as model pruning, quantisation, and adaptive communication strategies are expected to play a crucial role in achieving efficient federated anomaly detection. These advancements will enable practical deployment in large scale IoT environments.

7.4 Integration with Advanced AI and Adaptive Learning Techniques

The integration of federated learning with advanced artificial intelligence techniques presents a promising direction for enhancing anomaly detection capabilities. Hybrid models that combine deep learning with reinforcement learning, transfer learning, or meta learning can improve adaptability and performance in dynamic environments [10, 11].Adaptive learning strategies that dynamically adjust to changing data distributions and network conditions are also gaining importance. These approaches enable federated systems to continuously learn from evolving data patterns and maintain consistent performance over time. Furthermore, the combination of federated learning with emerging technologies such as edge intelligence and next generation communication networks is expected to further enhance scalability and efficiency.

Overall, these future directions highlight the potential for significant advancements in federated anomaly detection systems. Continued research and innovation in these areas will play a vital role in developing secure, scalable, and intelligent IoT systems capable of addressing the growing challenges of modern distributed environments.Another emerging direction involves the use of collaborative multi model systems, where multiple federated models operate simultaneously to address different types of anomalies. Such systems can improve detection coverage and adaptability in complex IoT environments. Continued research in this area is expected to significantly enhance the robustness and flexibility of federated anomaly detection frameworks.

8 Conclusion

Out in the open, federated anomaly detection tackles security issues across scattered IoT setups. Instead of moving data, models learn together while staying local. This keeps private information protected yet still sharp at spotting threats. With smarter algorithms now linked to distributed updates, catching unusual patterns in vast mixed device networks works far better than before. Away from old central hubs, these methods handle growth more smoothly, cut down transmission load, while locking down sensitive inputs tighter.

Looking at different ways to detect anomalies in federated systems, the research explored how models are combined, structured through layered networks, or secured against threats. While some designs rely on centralised coordination, others distribute control more evenly among nodes, each affecting accuracy differently. Optimisation plays a key role; better results often come from fine tuned updates rather than raw data volume. Surprisingly, resilience emerges not just from strong algorithms but also from how they adapt during operation. Real tests in healthcare, manufacturing, and smart cities show these systems work beyond theory handling noise, delays, and partial failures without collapsing. Performance gains appear when learning adjusts dynamically instead of following fixed rules. Not every method scales well, yet certain configurations manage complexity by balancing local autonomy with global consistency. Effectiveness depends heavily on context; one size rarely fits all. Still, patterns emerge: stability increases when trust mechanisms guard model exchanges. Across cases, success links closely to how smoothly devices collaborate despite limited visibility into each other's processes.

Even with progress, problems persist, like managing uneven data splits, cutting down on message traffic between devices, while staying strong under targeted manipulation attempts. Tackling such hurdles leans heavily on ongoing work in learner models, safer ways to combine updates, along with networked training setups that do not rely on a central hub. Progress here should lift both dependability and room to grow within distributed outlier spotting tools, fitting well into future internet-connected device ecosystems.

Growing use of IoT tools in areas like health services, transit systems, and urban infrastructure underlines the need for reliable anomaly detection methods. Because data stays on local devices, federated learning supports both scalability and user privacy. With ongoing progress, linking this approach to edge processing, machine learning models, and next gen network setups could enhance how well these systems function. One outcome may be faster responses without sacrificing security. Still, real-world conditions will shape whether such combinations deliver consistent results.

Ultimately, using federated learning for spotting unusual behaviour improves security across scattered IoT setups. Because it balances private data handling with growing system demands and smart analysis, it fits well within today's networked technologies. Progress in these methods will quietly influence how strong and smooth future device environments become. A shift toward this model should support stronger, smarter networks across connected devices.

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